

Research paper

Multi-objective optimization and feasibility analysis of integrated biogas–solar energy systems for rural electrification

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ABSTRACT

The growing demand for sustainable electricity in emerging economies necessitates hybrid systems that leverage local renewable resources while remaining economically viable. This study optimizes and evaluates photovoltaic–biogas (PV–BG) hybrid systems for Rosso, Mauritania, through a techno-economic and environmental framework. HOMER Pro was used for baseline modeling, while Grey Wolf Optimizer (GWO) and Whale Optimization Algorithm (WOA) refined both on-grid and off-grid designs. The optimal on-grid configuration—801 kW PV, 100 kW BG generator, and 408 kW converter—achieved a Levelized Cost of Energy (LCOE) of \$0.041/kWh, Net Present Cost (NPC) of \$1.89 M, Payback Period (PP) of 6.6 years, Internal Rate of Return (IRR) of 14%, and Return on Investment (ROI) of 11%. GWO and WOA further reduced LCOE to \$0.038/kWh and \$0.036/kWh and NPC to \$1.81 M and \$1.77 M, shortening PP to 6.4 and 6.1 years. Environmental analysis showed an annual offset of 1,220 tCO₂ and a 100% renewable fraction. The results provide a scalable framework for hybrid energy planning, supporting policy development and investment strategies toward low-carbon power systems.

1. Introduction

Fuel and energy shortages have become pressing issues in the twenty-first century, with growing global demand coinciding with the depletion of fossil fuel reserves [1,2]. Heavy dependence on fossil fuels has further intensified climate change, air pollution, and ecosystem degradation [3,4]. Consequently, international frameworks have been established to

phase out or significantly reduce fossil fuel consumption in the coming decades [5–7]. The 2015 Paris Agreement emphasizes the urgent need to limit global temperature rise to 1.5 °C above pre-industrial levels to mitigate severe consequences, including extreme weather events, rising sea levels, biodiversity loss, and threats to human health [8]. Amid these challenges, renewable energy sources have emerged as viable solutions to mitigate environmental impacts and enhance energy

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Nomenclature

PV	Photovoltaic	$CT_{pv,STC}$	PV cell temperature under STC..... °C
HOMER	Hybrid Optimization of Multiple Renewable Resources	$E_{PV}(t)$	Energy generated by PV array at time t kWh
LCOE	Levelized Cost of Energy	$G(t)$	Hourly solar irradiance at time t kWh/m ²
ROI	Return on Investment	η_{pv}	PV module efficiency..... %
NPC	Net Present Cost	AR	Total surface area of PV panels..... m ²
CO ₂	Carbon Dioxide	P_{cv}	Converter rated capacity..... kW
HRES	Hybrid Renewable Energy Sources	$E_{D(max)}$	Maximum energy demand..... kWh
LPG	Liquid Petroleum Gas	η	Converter efficiency..... %
GWO	Grey Wolf Optimizer	$E_{ac-dc}(t)$	Energy converted from AC to DC at time t kWh
WOA	Whale Optimization Algorithm	$E_{dc-ac}(t)$	Energy converted from DC to AC at time t kWh
STC	Standard Temperature Condition	η_{Rec}	Rectifier efficiency..... %
IRR	Internal Rate of Return	η_{Inv}	Inverter efficiency..... %
BEP	Break-Even Point	E_{Gload}	Energy supplied by battery or grid to meet load kWh
PP	Payback Period	E_{BPE}	Surplus energy generated by biogas generator..... kWh
GHG	Greenhouse Gas	P_{BG}	Power output from biogas generator..... kW
O&M	Operation and Maintenance	B_A	Biomass availability..... kg/hr
RF	Renewable Fraction	CV_b	Calorific value of biogas..... kJ/kg
MCDM	Multi-Criterion Decision-Making	η_g	Efficiency of biogas generator..... %
MHRES	Multi-Criterion Decision-Making Approach to Assess Optimal Micro-Hybrid Renewable Energy Systems	η_{gas}	Gasification efficiency of gasifier..... %
P_{PV}	PV array output power..... kW	NP_C	Net Present Cost..... \$
R_{pv}	Rated capacity of the PV array..... kW	NC_i	Net cash flow in year i \$
DF_{pv}	PV module derating factor..... %	R	Discount rate..... %
S_{IR}	Solar irradiance incident on PV array..... kW/m ²	$TInvC$	Total investment cost..... \$
$S_{IR,STC}$	Solar irradiance at Standard Test Conditions..... kW/m ²	IRR	Internal Rate of Return..... %
α_T	Temperature coefficient of power..... %/°C	BEP	Break-Even Point..... years
CT_{pv}	PV cell temperature at given time..... °C	CF	Annual cash flow..... \$

security [9–11]. Among them, Solar energy and biogas stands out as a sustainable option due to its dual role in renewable energy production and organic waste management [12,13]. PV systems convert solar radiation directly into electrical energy through the photovoltaic effect, primarily employing semiconductor materials such as crystalline silicon or thin-film technologies. A typical PV system consists of solar modules, power inverters, mounting structures, and associated balance-of-system components that together determine overall system performance and reliability [14]. Advances in PV technology focus on enhancing conversion efficiency, reducing costs, and improving durability to support integration within hybrid renewable energy systems. The integration of PV with complementary resources, such as biogas, leverages the synergistic potential of diversified generation, thereby enhancing system reliability and economic viability [15,16]. On the other hand, derived from the anaerobic digestion of agricultural, municipal, and industrial waste, biogas is a carbon-neutral fuel that reduces reliance on fossil fuels, curbs greenhouse gas (GHG) emissions, and supports rural development by creating jobs and closing the nitrogen cycle [17,18]. By converting organic residues into usable energy, biogas systems not only provide a reliable renewable source but also address spatial environmental concerns, such as deforestation and waste pollution. This dual function of energy generation and waste management is particularly valuable in countries with abundant agricultural and organic waste streams. Fig. 1 illustrates the top ten countries with the highest biogas generation potential, highlighting the global relevance of this renewable resource.

Fig. 2 illustrates the remarkable global increase in biogas production from the early 2000s to 2020, a period that witnessed a sharp rise in biogas-based energy generation. In developing regions, particularly across Asia and Africa, biogas programs have gained substantial traction. As shown in Table 1, countries like China and India pioneered large-scale biogas development programs as early as the 1970s, leading to significant deployment milestones. China has installed approximately 35 million biogas plants, while India has developed over 4.5 million units. Other nations, including Nepal, Vietnam, and Bangladesh, have

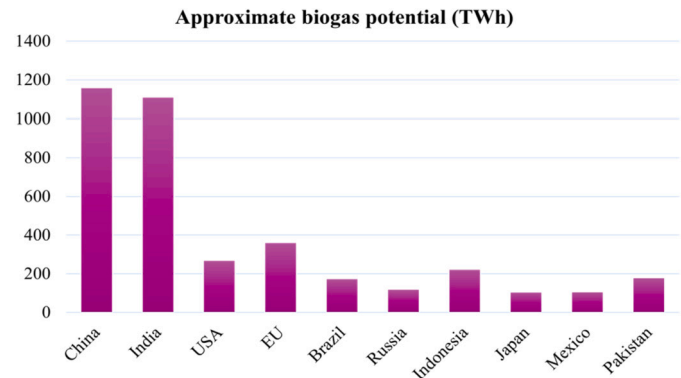


Fig. 1. Top 10 countries with the highest biogas generation potential [19].

also implemented successful initiatives, demonstrating the widespread adoption of biogas technologies in regions with abundant organic resources and growing energy demands.

A growing body of literature underscores the potential of biogas-to-electricity systems as a renewable energy solution with broad applicability. Agyenim *et al.* studied the feasibility of a hybrid power system in Ghana, combining biogas and solar PV energy. Their study found that integrating solar PV with existing biogas plants is technically viable and cost-effective, enhancing daily power production, using a multi-criteria analysis method to evaluate three renewable energy business models [22]. Odoi-Yorke *et al.* investigated the potential of a solar PV/biogas/battery hybrid energy system in Ghana, aiming to deliver electricity to remote communities at a cost-effective LCOE while reducing GHG emissions. Using HOMER software, they evaluated technical, economic, and environmental performance, reporting a cost-efficiency of approximately \$0.256/kWh [23]. Dodo *et al.* examined the technical and financial feasibility of a solar PV/biogas/battery hybrid system using municipal solid waste as feedstock for a biogas generator in a remote

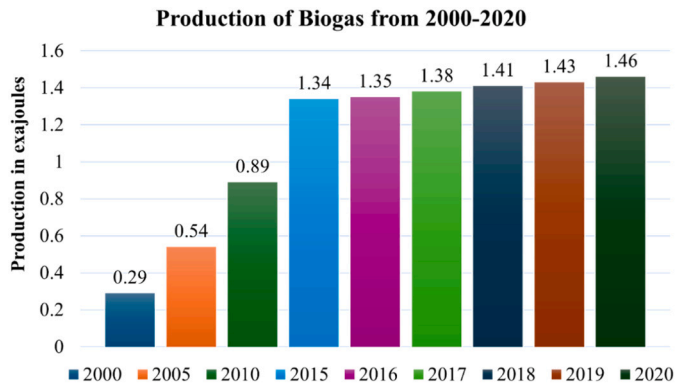


Fig. 2. Production of biogas in the world from 2000 to 2020 [20].

Table 1

Installed Biogas Plants in Developing Countries, Particularly in Asia and Africa [21].

Country	Year of Program Initiation	Cumulative Biogas Plants Installed
China	1974	35,000,000
India	1970s	4,500,000
Nepal	1992	268,464
Vietnam	2003	152,349
Bangladesh	2006	26,311
Cambodia	2006	19,173
Indonesia	2009	7,835
Pakistan	2009	2,324
Bhutan	2011	265
Rwanda	2007	2,619
Ethiopia	2008	5,011
Tanzania	2008	498
Kenya	2009	6,749
Uganda	2009	3,083
Burkina Faso	2009	2,013
Cameroon	2009	159
Benin	2010	42
Senegal	2010	334
Rwanda	2018	2,400

Nigerian village. The optimized system, modeled with HOMER, met the village's energy demand at an LCOE of \$0.2917/kWh, significantly reducing GHG emissions and providing a sustainable energy solution [24]. Likonge Makai and Olawale Popoola applied a Multi-Criterion Decision-Making (MCDM) approach to assess optimal Micro-Hybrid Renewable Energy Systems (MHRES) for rural Zambia. Their research identified a biogas-solar PV system with SBS 190 battery storage as the most suitable option for replacing fossil fuels, confirmed through multiple validation methods [25]. Ennemiri *et al.* evaluated a hybrid solar PV/biogas/battery system for a commercial platform in Berkane, Morocco. The optimized configuration, modeled using HOMER, demonstrated a lower LCOE of \$0.280/kWh and a 40% reduction in CO_2 emissions compared to a biogas-only system, with a sizing of 231 kW PV, 170 kW biogas, and 201 kWh Li-ion battery capacity [26]. Jumare *et al.* analyzed a grid-connected PV/wind/biogas hybrid system in northern Nigeria using HOMER, Excel, and GaBi tools. Their findings revealed a 3% increase in energy supply and substantial cost reductions for the grid-connected configuration, including a 68% lower net present cost (NPC) and 85% lower LCOE than its off-grid counterpart, along with major reductions in emissions and component sizing [27]. Al-Najjar *et al.* focused on optimizing an HRES combining biomass and solar PV for 94 residential buildings with a demand of 84.5 kWp. Using HOMER Pro and the IEC 61724 standard, they assessed the influence of solar irradiation, efficiencies, and fuel consumption on system performance and costs. Their study evaluated energy efficiency, system sizing, and economic viability as key performance indicators [28]. Despite these contributions, several studies still struggle to fully meet the electricity demands of remote re-

gions solely through renewable sources or face feasibility constraints due to high generation costs and overall project expenses [29–32]. (See Table 2.)

These studies demonstrate that hybrid renewable energy systems incorporating biogas can deliver substantial benefits in terms of cost-effectiveness, emission reductions, and operational efficiency. However, a notable research gap emerges from the reviewed literature and tabular comparison:

- Most existing studies predominantly rely on either HOMER Pro software or optimization algorithm for system modeling and optimization, often without benchmarking against alternative or advanced metaheuristic algorithms, limiting optimization potential.
- Few investigations comprehensively compare on-grid versus off-grid configurations within the same study, which is essential for context-specific feasibility assessments.
- There is limited consideration of multi-objective optimization that simultaneously integrates technical, economic, and environmental performance indicators in a unified framework.
- Regional studies rarely focus on West African contexts, especially Mauritania, where renewable resource potential is significant but under-explored in hybrid system design.

This study addresses these gaps by applying advanced optimization algorithms alongside HOMER, evaluating both on-grid and off-grid configurations, and focusing on Mauritania's unique energy landscape to propose feasible, sustainable hybrid biogas–PV systems. Mauritania, located in West Africa, remains heavily dependent on imported fossil fuels, and rapid urbanization coupled With population growth has significantly increased energy demand, raising pressing concerns about sustainability and economic stability. With 58.6% of the population living in poverty and unable to afford reliable electricity [51], transitioning from traditional wood burning to biogas offers a cleaner, more sustainable energy alternative [52]. Fig. 3 illustrates the biogas generation potential across all provinces of Mauritania.

The substantial livestock population in Mauritania exceeds 29.5 million heads, with the livestock sector expanding at a rate of more than 3.5% annually, underscoring the significant potential of biogas production to support sustainable energy solutions [53]. Leveraging these abundant livestock resources, large-scale biogas facilities could offer a competitive alternative to liquid petroleum gas (LPG), generating an annual production capacity of 2,125 million cubic meters, and potentially reaching 2,451 million cubic meters per year when accounting for all waste types [54]. As of 2023, Mauritania has made notable strides in improving electricity access; however, disparities between urban and rural regions remain stark. National electricity coverage stands at approximately 47%, with urban areas achieving around 83% access, while only 7% of rural households have electricity [55,56].

In this context, the overarching aim of this study is to assess the feasibility and optimization of integrating biogas and PV systems for electricity generation in Mauritania, offering a sustainable pathway to mitigate the nation's energy access challenges. The specific objectives of this research are:

1. Assess the technical and economic viability of a hybrid biogas–PV power system in Mauritania by analyzing local energy demand, resource availability, and climatic conditions.
2. Employ HOMER software to evaluate and compare on-grid and off-grid biogas–PV system configurations.
3. Integrate advanced optimization algorithms to benchmark and enhance HOMER outputs, identifying the most cost-effective and efficient system design beyond traditional HOMER-based approaches.
4. Develop a structured framework for designing and implementing biogas–PV hybrid systems, detailing key sizing criteria, component specifications, performance indicators, and operational guidelines.

Table 2

Comparison of related works in hybrid renewable energy systems incorporating biogas.

Ref.	Year	Region	Optimization Algorithm	System Configuration	Financial Viability Metrics	Main Results	Limitations
[33]	2024	Ghana (Ashanti)	RETSCREEN + Excel	PV + AD + Pyrolysis	NPV = €13M, Payback = 12 yrs, LCOE = 0.3–0.68 €/kWh	Electricity-fuels option most viable; fuels-only also beneficial	High LCOE; location-specific; limited optimization
[34]	2025	Kenya (Kesses)	HOMER	PV 25.5 kW + BG 3 kW + Battery 143 kWh + Converter 18.2 kW	LCOE = 0.171 \$/kWh, CO ₂ saved = 60,193 kg/yr	Rural electrification feasible	Location-specific; no advanced optimization
[35]	2020	Pakistan	–	PV 2.84 kW + Biogas (4 m ³)	–	Demonstrated feasibility at small scale; outputs: 2.14 kW (noon)	Short testing; no techno-economic optimization
[36]	2024	Pakistan (Tangi)	HOMER	PV 91.4 kWp + Hydro 19.6 kW + BG 50 kW + 36 Batteries	LCOE = 0.0728 \$/kWh; Annual prod. = 294,782 kWh	Low GHG emissions; viable hybrid	Region-specific; limited scalability; single optimizer
[37]	2025	Turkey (Hatay)	HOMER Pro	PV 200 kW + BG 100 kW + Conv. 72 kW + Electrolyzer 120 kW	COE = 0.2094 \$/kWh; NPC = 1.92 M\$	CO ₂ emissions reduced by 95.78%	Location-specific; scalability concerns
[38]	2024	–	ML-based methane forecasting	PV + BG (beef manure feedstock)	–	Forecast error reduced from 12.82% to 8.28%	No techno-economic analysis
[39]	2021	India (Assam)	SAM model	PV 1 kW + Biogas 3.5 kVA (battery backup)	NPV = \$1562; LCOE = 0.21 \$/kWh; CO ₂ mitigation = 104.6 t	Economic viability over 25 yrs; CO ₂ valued at \$2090	Small scale; no grid-integration scenarios
[40]	2022	Jordan (Amman)	MOFEPSO + fuzzy logic	On-grid MW-scale PV + BG	TNPC = 64.5 MUSD/yr; IR = 96.05%	Multi-objective optimization successful	Very large-scale; high cost; limited emission analysis
[41]	2023	Egypt (Ataka)	HBO, GWO, PSO	Standalone PV–wind–biomass–hydro storage	COE = 0.275 \$/kWh; NPC = 8.06 MUSD	HBO best performer; meteorological data-based	No sensitivity analysis; limited LCA
[42]	2022	Egypt (Alrashda)	QRUN (vs RUN, AO, GWO)	PV–biomass gasifier–battery	COE = 0.0801 \$/kWh; NPC = 3.35 M\$	QRUN yielded best cost–reliability trade-off	Village-scale only
[43]	2023	Spain (Valencian Community)	Demand–price curve-based	Biomass BG 140 kW + PV 80 kW (grid-tied)	IRR = 6.13% over 12 yrs	Optimal for rural community	Mid-scale only; no environmental analysis
[44]	2022	China (Xuzhou)	HOMER	PV 400 kWp + BG 100 kW + Diesel + Battery	COE = 0.24 \$/kWh; NPC = 1.81 M\$; CO ₂ saved = 1.3 Mt	Optimal PV/BG/battery mix	Sensitivity analysis on BG and battery
[45]	2024	Morocco (Berkane)	HOMER	PV 231 kW + BG 170 kW + Conv. 140 kW	LCOE = 0.280 \$/kWh; CO ₂ ↓ 40%	Optimal commercial sizing	Platform-specific; limited scale
[46]	2023	India (West Bengal)	DGWO	PV/biomass/battery hybrid	NPC = \$159,649; LCOE = \$0.0712/kWh	Scenario-2 gave best techno-economics	Social + technical constraints
[47]	2025	Cameroon (Douala)	HOMER	PV + BG + Diesel + Battery (grid-tied)	LCOE = \$0.0756/kWh; NPC = \$300,661; Payback = 4.96 yrs	CO ₂ reduced 50%; viable for institutions	Focused on single site
[48]	2025	Bangladesh	HOMER	PV + BG + Wind + H ₂ (transport)	Cost = \$0.0141/km (Model 1); 99.5% CO ₂ reduction (Model 3)	Net-metering EVs vs H ₂ vehicles compared	Context-specific; model-based only
[49]	2024	Türkiye (Central Anatolia)	HOMER	Biogas + PV + Wind (grid-tied farm)	LCOE = 0.0688 \$/kWh; NPC = \$70,777; 89.9% RE fraction	Viable farm-level hybrid	Sensitivity on load/grid sellback
[50]	2024	Brazil (Pará)	–	PV–BG hybrid (cassava wastewater CHP)	LCOE = 0.041–0.045 \$/kWh; NPV = \$321–339k; IRR = 35–50%; Payback = 3–5.5 yrs	High IRR; surplus sales improve economics	Case-specific; CHP inclusion

5. Formulate strategic recommendations for policymakers, stakeholders, and investors to facilitate large-scale deployment and adoption of biogas–PV systems as a sustainable electricity solution for Mauritania.

The key novelties of this study are:

1. The integration of GWO and WOA to refine system design and economic performance, offering a comparative benchmark against HOMER's optimization outcomes.
2. The first comprehensive feasibility study of biogas–PV hybrid systems in Mauritania, explicitly factoring in local resource potential, economic limitations, and climate-specific constraints.
3. A systematic trade-off analysis between on-grid and off-grid configurations, examining cost efficiency, environmental impact, and system resilience to inform region-specific decision-making.
4. A multi-objective evaluation combining economic metrics (LCOE, NPC, IRR, ROI, payback period) with environmental indicators (GHG reduction, renewable fraction), creating a robust decision-making framework for policy and investment.

In summary, the motivation for this study arises from the urgent need to provide sustainable, affordable, and reliable electricity to un-

derserved communities by leveraging locally available renewable resources. A review of existing literature reveals that while numerous studies have demonstrated the benefits of hybrid renewable energy systems, particularly those incorporating biogas, most have relied solely on HOMER-based optimization, with limited exploration of advanced meta-heuristic techniques. This highlights a clear research gap in combining traditional techno-economic modeling with advanced optimization algorithms for hybrid PV–biogas systems under both on-grid and off-grid scenarios. The primary contributions of this work are threefold: (i) integrating Grey Wolf Optimizer (GWO) and Whale Optimization Algorithm (WOA) alongside HOMER to enhance system design and economic performance; (ii) providing a comparative assessment of on-grid and off-grid PV–biogas configurations under region-specific resource and economic constraints; and (iii) delivering a comprehensive multi-criteria evaluation framework that jointly considers economic indicators (LCOE, NPC, IRR, ROI, payback period) and environmental impacts (GHG reduction, renewable fraction) to inform policy and investment decisions.

The remainder of this paper is organized as follows: Methodology section presents the research methodology, including data collection, resource assessment, simulation and optimization setup, technical specifications of the HRES components, as well as the economic and environmental modeling frameworks. Results Section reports the on-grid and off-grid optimization results, followed by a comparative analysis

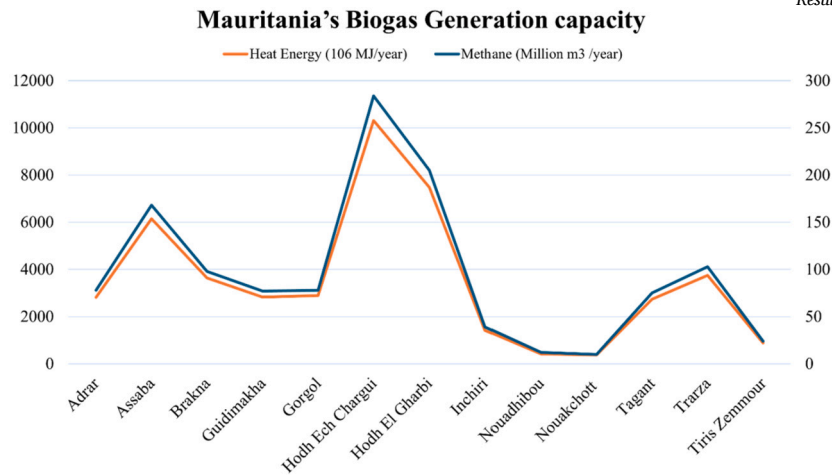


Fig. 3. Data about the possibility of producing electricity from animal feces in several Mauritanian areas [18].

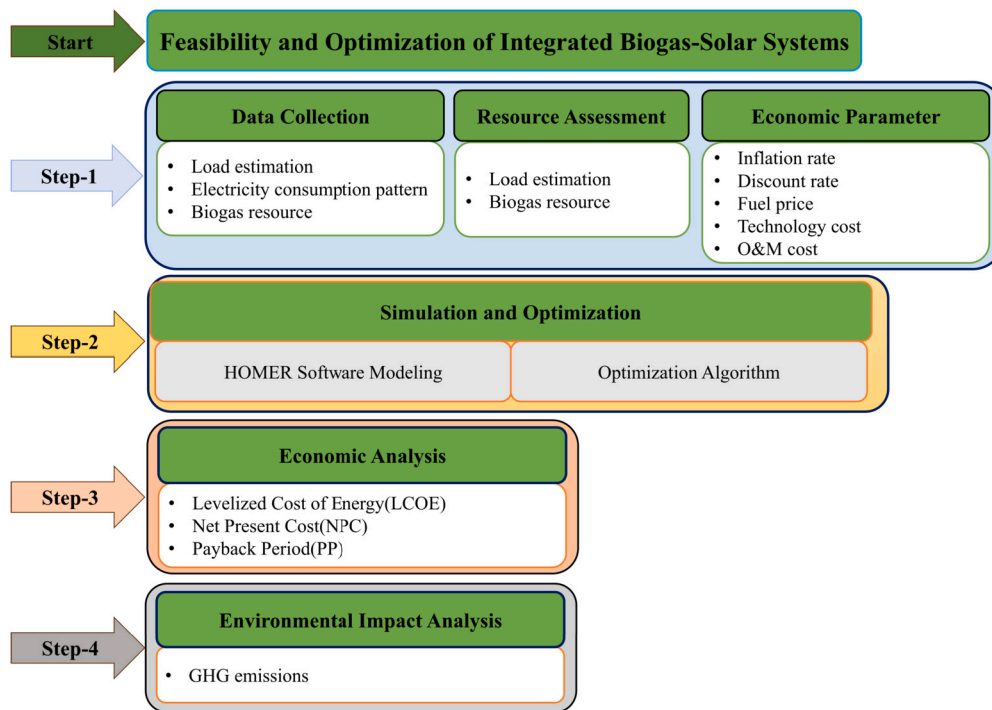


Fig. 4. Methodology flowchart of the feasibility and optimization of integrated PV-Biogas system study.

with similar studies in the literature. The discussion section discusses the implications of the findings, highlighting technical, economic, and environmental perspectives. Finally, the conclusion section summarizes the key conclusions of the study and offers recommendations for future research and large-scale implementation of hybrid renewable energy systems.

2. Methodology

The methodology of this study was structured to comprehensively achieve the stated objectives and is summarized in Fig. 4. The process integrates data collection, resource assessment, system simulation, advanced optimization, feasibility analysis, and environmental evaluation, ultimately leading to actionable policy recommendations.

- i. **Data Collection:** Surveys were conducted to collect household and agricultural load profile data, capturing electricity consumption patterns across selected regions of Mauritania. In addition, informa-

tion on local biogas resources—including organic waste availability and potential feed-stock sources was gathered to support biogas potential estimation.

- ii. **Resource Assessment:** A comprehensive assessment of biogas and solar energy resources was performed. Data on agricultural residues, livestock manure, and municipal solid waste were analyzed to determine the potential for biogas production, while solar irradiance data were examined to evaluate regional solar energy availability.
- iii. **Simulation and Optimization:** The HOMER software was utilized to model multiple PV–biogas hybrid configurations. Mauritania-specific inputs—including solar irradiance, biogas production potential, and system cost structures—were incorporated. Both on-grid and off-grid scenarios were evaluated under variations in system capacity, storage options, and economic assumptions. To enhance the optimization process, two advanced metaheuristic algorithms, GWO and WOA, were applied to benchmark and validate HOMER-generated results.

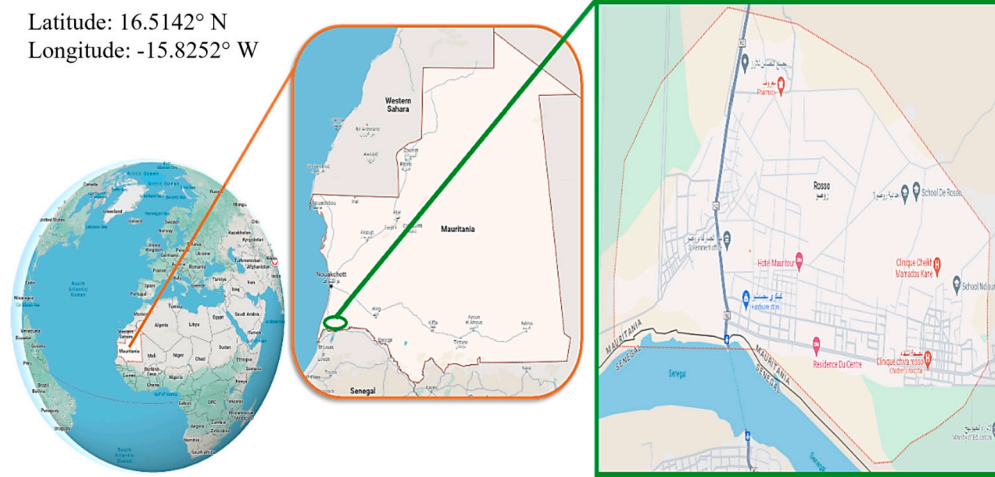


Fig. 5. Geographical location of Rosso, Mauritania (Latitude 16.5142° N, Longitude 15.8252° W).

Table 3
Number and type of surveyed structures in Rosso.

Infrastructure Type	Number of Structures
School	2
Household	120
Mosque	4
Shop	10
Farms	4
Government Office	1
Hospital	1
Total	142

- iv. **Feasibility Analysis:** A detailed feasibility assessment was performed to evaluate the technical and economic performance of the hybrid systems. Key performance metrics, including the LCOE, PP, and system reliability, were examined under various operational scenarios.
- v. **Environmental Impact Assessment:** The study quantified the potential reduction of GHG emissions achieved by the hybrid systems compared to conventional fossil fuel-based generation.
- vi. **Policy Recommendations:** Based on the technical, economic, and environmental findings, policy and investment recommendations were formulated.

Novelty within the methodology of this study is, it introduces the integration of advanced optimization techniques-specifically GWO and WOA-to refine the performance and cost-effectiveness of PV-biogas hybrid systems. Unlike conventional HOMER-only optimization approaches, the use of these algorithms enables exploration of alternative system configurations, enhancing economic outcomes while maintaining operational reliability. Furthermore, the dual evaluation of both on-grid and off-grid scenarios tailored to Mauritania's resource and infrastructure landscape provides an unprecedented comparative perspective. The methodological framework also emphasizes localized biogas resource assessment by leveraging livestock waste data to ensure realistic and scalable hybrid energy solutions.

2.1. Data collection and load profile construction

The process of data collection and load profile construction represents a important step in assessing the feasibility of the PV-Biogas hybrid system for the case study area, Rosso, Mauritania. The site is geographically located at precise coordinates of latitude 16.5142° N and longitude 15.8252° W, which ensures accuracy in solar irradiance estimation, climatic data acquisition, and load demand characterization. Accurate and

Table 4
Daily load calculations for Rosso, Mauritania.

Load Type	No. of Loads	Avg. Power (W)	Hours	Energy (kWh)
Light	307	50	10	153.5
Fan	221	70	10	154.7
TV	24	100	10	24.0
Refrigerator	15	1000	5	75.0
Streetlight	25	120	12	36.0
Pump	15	1500	7	157.5
Total				600.7

detailed load profiles are essential to understand electricity demand patterns and to optimize the design of renewable energy systems. This study was conducted in Rosso, a rural city located in the southwestern region of Mauritania near the Senegal River. Rosso is a major agricultural hub and serves as an important link between Mauritania and Senegal. The geographical position of Rosso, shown in Fig. 5, highlights its strategic relevance for this study, particularly in addressing the energy needs of both urban and rural communities. A structured field survey was conducted to gather energy consumption data from a variety of infrastructures, including residential, commercial, educational, agricultural, and institutional facilities. The surveyed infrastructures, summarized in Table 3, provide a holistic representation of electricity usage in Rosso. In total, 142 infrastructures were surveyed, including 120 households, 10 shops, 4 mosques, 4 farms, 2 schools, 1 hospital, and 1 government office. This diverse sample provides a comprehensive understanding of electricity consumption across different sectors in Rosso. Daily load calculations, provided in Table 4, detail the electricity consumption by type of load. Lighting contributes 153.5 kWh/day, fans 154.7 kWh/day, and pumps 157.5 kWh/day, making pumps and fans the largest consumers of electricity. Refrigerators, televisions, and streetlights contribute 75 kWh/day, 24 kWh/day, and 36 kWh/day, respectively. In total, the surveyed infrastructure consumes 600.7 kWh per day. This detailed load profile offers an accurate representation of demand patterns, enabling optimized system sizing and component selection for the PV-Biogas hybrid systems.

2.2. Resource assessment

A thorough resource assessment is essential for determining the feasibility of hybrid biogas-PV systems [58]. This section evaluates the solar and biomass resources available in the region, as these inputs are fundamental for ensuring the long-term sustainability and performance of the proposed hybrid system.

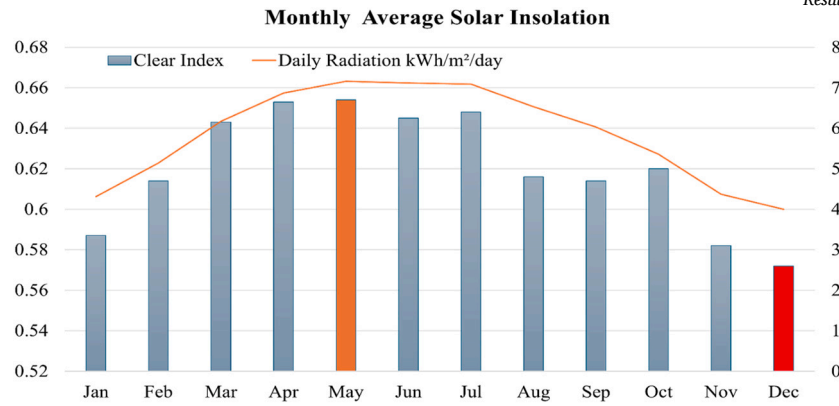


Fig. 6. Monthly average solar radiation and clearness index for Rosso, Mauritania [57,13].

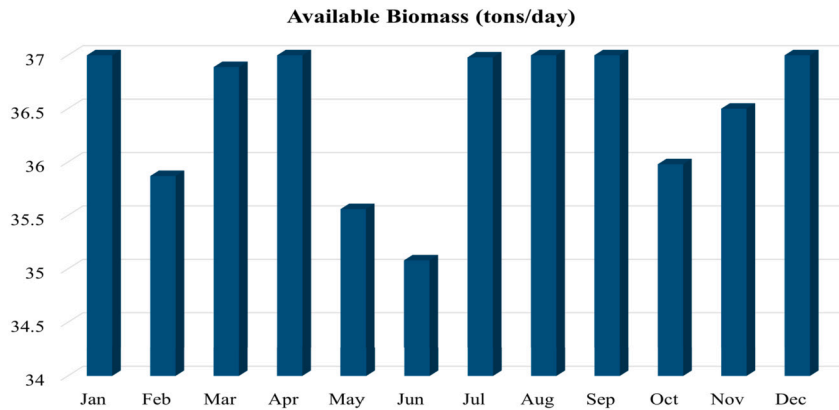


Fig. 7. Monthly average biomass availability in Rosso, Mauritania [13,54].

2.2.1. Solar resources in Mauritania

Fig. 6 shows monthly average solar radiation for Rosso, with clearness index ranging from 0.572 to 0.654. Radiation rises from 4.31 kWh/m²/day in January (CI 0.587) to a peak of 7.16 kWh/m²/day in May (CI 0.654), remaining high through July. It then declines to 4.0 kWh/m²/day in December (CI 0.572). The consistently high irradiance, particularly from March to July, highlights Rosso's strong potential for reliable and sustainable PV deployment.

2.2.2. Biomass resources in Mauritania

Fig. 7 presents the monthly average biomass availability in Rosso, measured in tons per day [54]. Biomass levels remain stable throughout the year, averaging approximately 37 tons/day. The lowest availability is observed in June (35.08 tons/day), while the highest, consistently at 37 tons/day, occurs in January, April, August, September, and December. This stability indicates a reliable and predictable feedstock for biogas production, ensuring steady supply for hybrid system operations. The continuous availability of biomass enhances the feasibility of integrating biogas production with PV systems, contributing to year-round electricity generation and energy security.

While multi-technology hybrid systems incorporating components such as wind turbines (WT), diesel generators (DG), battery energy storage systems (BESS), fuel cells (FC), electrolyzers (ELZ), and hydrogen tanks (HT) have demonstrated advantages in certain contexts, the present study focuses specifically on a PV–biogas hybrid configuration. This selection is motivated by three key factors: (i) resource availability—Mauritania possesses abundant solar irradiance and significant untapped biomass potential from agricultural and livestock residues, whereas wind and hydrogen infrastructure remain limited; (ii) economic and logistical feasibility—PV and biogas technologies have comparatively lower capital costs, reduced technical complexity, and greater accessibility for rural deployment; and (iii) policy alignment—

current national energy strategies emphasize the development of solar and biomass resources to reduce dependence on imported fossil fuels. Narrowing the scope to PV–biogas allows for a detailed techno-economic and environmental analysis tailored to local conditions, while still providing a methodological framework that can be extended in future studies to incorporate additional renewable and storage technologies.

2.3. Simulation and optimization setup

2.3.1. HOMER simulation

The simulation setup outlines the process and parameters used to model and analyze the feasibility of integrating biogas and PV systems for electricity generation in Rosso, Mauritania. HOMER software was employed to simulate multiple system configurations and evaluate their performance, cost-effectiveness, and environmental impact. HOMER is widely recognized in the research community for its robust modeling and optimization capabilities for hybrid renewable energy systems, making it an appropriate tool for this study [59]. Key input parameters for the simulation included solar irradiance data, biomass availability, surveyed household and agricultural load profiles, and economic factors such as installation costs, O&M expenses, and fuel prices. Separate scenarios for on-grid and off-grid systems were developed by varying capacities, storage solutions, and economic assumptions.

Fig. 8 illustrates the configurations for both on-grid and off-grid systems modeled in HOMER. The on-grid configuration integrates a biogas generator, PV array, power converter, battery storage, and a grid connection. This setup enables excess electricity to be exported to the grid when production exceeds demand. In contrast, the off-grid system configuration, which excludes a grid connection, is composed of a biogas generator, PV array, converter, and battery storage, designed to supply power to remote areas without centralized infrastructure. Both system

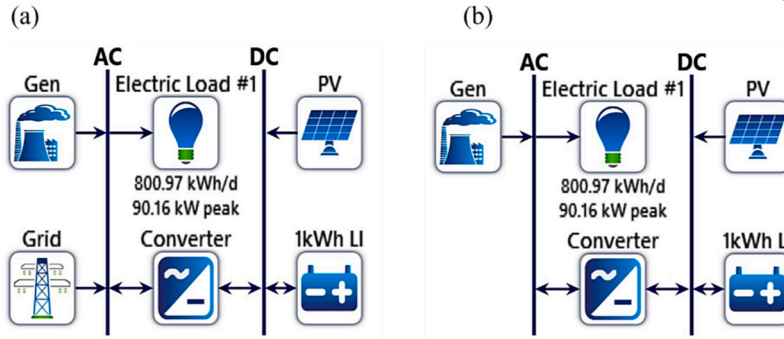


Fig. 8. System configurations simulated in HOMER: (a) On-grid system and (b) Off-grid system.

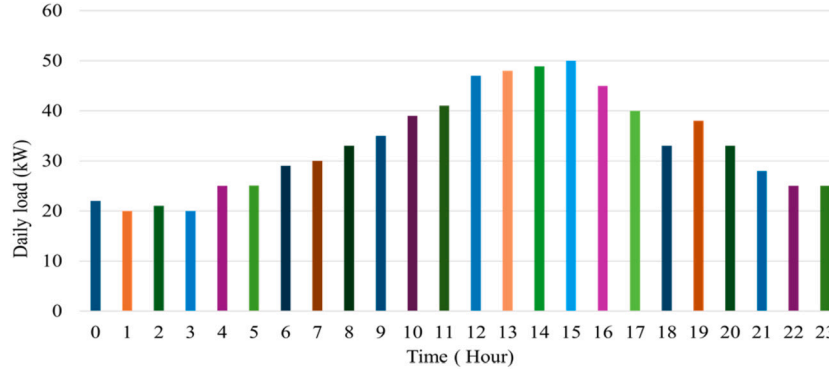


Fig. 9. Hourly load profile used in the HOMER optimization.

models were designed for an electricity load of 800.97 kWh/day, with a peak load of 90.16 kW. Notably, the simulated load was deliberately set 33% higher than the measured value to provide a buffer for demand fluctuations and ensure system resilience. This overestimation also allows the assessment of future scalability without major system redesign or additional investment in infrastructure.

Fig. 9 presents the hourly load profile applied in the HOMER model. The profile demonstrates typical daily variations in electricity demand, with lower consumption during early morning and late evening hours, and peak demand during the late morning and early afternoon. Incorporating this profile into the simulation enables accurate modeling of system operation, energy generation, storage, and distribution to reliably meet demand fluctuations throughout the day.

2.3.2. Optimization algorithms

In this study, two advanced optimization algorithms, the GWO and the WOA, were implemented to benchmark and compare the system configurations produced by the HOMER software. Both GWO and WOA are widely acknowledged for their superior exploration and exploitation mechanisms, which improve their capacity to identify near-global optimal solutions in complex, nonlinear search spaces [60,61]. These algorithms were chosen due to their proven effectiveness in addressing the non-linear, multi-dimensional optimization challenges inherent in hybrid renewable energy system design [62–64]. The optimal configuration obtained from HOMER served as the initial input for both GWO and WOA, enabling a direct comparison of the software's baseline optimization results with the more advanced algorithmic approaches. This ensured that all methods operated under consistent equations, constraints, and boundary conditions, providing a fair and valid basis for comparison [65].

Fig. 10 illustrates the overall process followed for system optimization using GWO and WOA. The flowchart outlines each key stage, beginning with input parameter initialization (e.g., PV capacity range, biogas generator size, storage options), followed by iterative exploration and exploitation phases. During each iteration, the algorithms evalu-

ate potential solutions using the defined objective functions—primarily minimizing the LCOE and NPC while ensuring a 100% renewable fraction. The process continues until convergence criteria are met, resulting in an optimal configuration of the hybrid biogas–PV system. By integrating GWO and WOA alongside HOMER, this study not only validates the HOMER results but also explores potentially superior configurations, thereby enhancing the methodological rigor and the reliability of system design recommendations.

2.4. Technical information about the components of the HRES

The proposed hybrid power system consists of PV panels, a biogas generator, a converter, and battery storage system. The biogas generator is connected to the AC bus, whereas the PV panels and battery storage system interface with the DC bus. A bi-directional converter mediates the interaction between the DC and AC buses, ensuring smooth energy transfer and operational coordination between components. This integrated configuration enables site-specific energy management and optimized power delivery to meet the demands of the Mauritania case study location.

2.4.1. PV panel

The market offers a wide variety of PV panels, with new models frequently introduced due to rapid technological advancements. In this study, the EVO 6N panel is selected for the hybrid system. The panel features an efficiency of 22.5% and a nominal power output of 700 W. It also exhibits a long operational lifespan and a low degradation rate. The detailed technical specifications of the selected PV panel are provided in Table 5.

The output power of the PV array is modeled using Equation (1) [67]:

$$P_{PV} = R_{pv} \times DF_{pv} \times \left(\frac{S_{IR}}{S_{IR,STC}} \right) \times [1 + \alpha_T (CT_{pv} - CT_{pv,STC})] \quad (1)$$

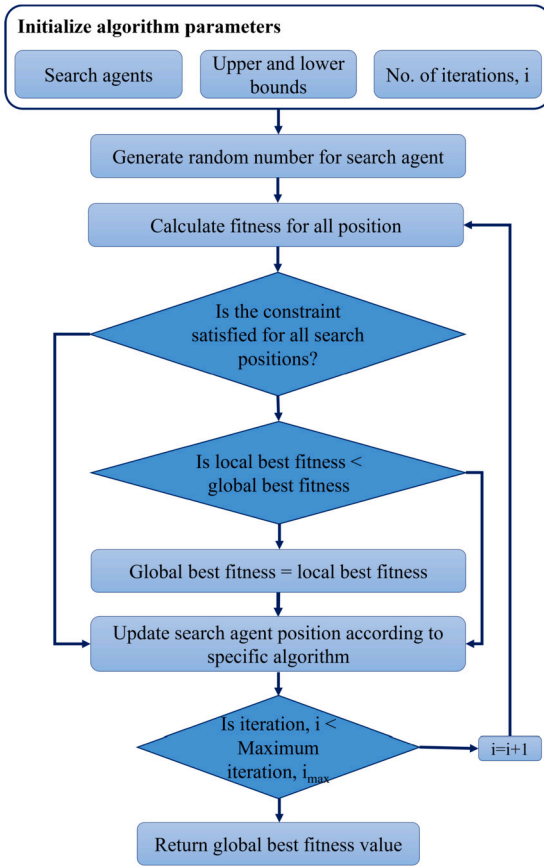


Fig. 10. General algorithm flowchart for system optimization using GWO and WOA.

Table 5

Technical specifications of EVO 6N PV panel [66].

Parameter	Value / Units
Manufacturer	SunEvo Solar
Model	EVO 6N SE6-66HBD
Panel dimension (L×W×H)	2384×1303×35 mm
Cell type	HJT Mono 210×105 mm
Panel weight	38.7 kg
Number of cells	132
Nominal power	700 W
Maximum Voltage	42.10 V
Open-circuit voltage (V_{oc})	50.13 V
Short-circuit current (I_{sc})	17.43 A
Panel efficiency	22.5%
Degradation Rate	0.6%
Warranty	30 years

where P_{PV} denotes the PV array output, R_{pv} is the PV array rated capacity (kW), DF_{pv} is the PV module derating factor (%), S_{IR} is the solar insolation incident on the PV array (kW/m^2), $S_{IR,STC}$ is the insolation at standard test conditions (STC) (kW/m^2), α_T is the temperature coefficient of power ($\%/^{\circ}\text{C}$), CT_{pv} is the PV cell temperature at a given time step ($^{\circ}\text{C}$), and $CT_{pv,STC}$ is the PV cell temperature under STC ($^{\circ}\text{C}$).

The energy production from the PV system is calculated using Equation (2) [68]:

$$E_{PV}(t) = G(t) \times \eta_{pv} \times AR \quad (2)$$

where $E_{PV}(t)$ represents the energy generated by the PV array (kWh), $G(t)$ denotes hourly solar irradiance (kWh/m^2), η_{pv} is the PV module efficiency (%), and AR is the total panel surface area (m^2).

2.4.2. Converter

The system employs a 1 kW on-grid bi-directional inverter, specifically the Canadian Solar CSI-1KTL1P-GI-FL model. This converter offers a lifespan of approximately 15 years and an efficiency of 97.2%. The rated capacity of the converter (P_{cv}) is determined using Equation (3) [62]:

$$P_{cv} = \frac{E_{D(max)}}{\eta} \quad (3)$$

where $E_{D(max)}$ represents the maximum energy demand (kWh) and η is the converter efficiency (%). The hourly energy exchange between AC and DC buses is represented by Equation (4):

$$E_{ac-dc}(t) = \eta_{Rec} \times E_{BPE}, \quad E_{dc-ac}(t) = \eta_{Inv} \times E_{Gload} \quad (4)$$

where $E_{ac-dc}(t)$ and $E_{dc-ac}(t)$ denote the hourly energy converted from AC to DC and DC to AC, respectively. η_{Inv} is the inverter efficiency, η_{Rec} is the rectifier efficiency, E_{Gload} refers to the hourly energy supplied by the battery or grid to meet the load, and E_{BPE} represents the surplus energy generated by the biogas generator.

2.4.3. Battery

The battery storage system serves as an essential component within the HRES, ensuring energy reliability and operational stability. During periods of surplus PV generation, the battery stores excess energy, while in the absence of sufficient PV power, it discharges to supply the deficit load demand. In this study, Lithium-ion (Li-ion) batteries are employed due to their superior energy density, long cycle life, and high round-trip efficiency. The nominal capacity of each battery unit is 1 kWh, with an overall round-trip efficiency of approximately 90%. The economic details of the battery system are summarized in Table 6.

To accurately capture the battery's performance, the following mathematical models are employed [69,70]:

Battery State of Charge (SOC) Update Equation

$$SOC(t) = SOC(t-1) + \frac{\eta_c P_{charge}(t) \Delta t}{C_{batt}} - \frac{P_{discharge}(t) \Delta t}{\eta_d C_{batt}} \quad (5)$$

where $SOC(t)$ = State of charge at time t , $SOC(t-1)$ = State of charge at previous time step, η_c = Battery charging efficiency (typically 90–95%), η_d = Battery discharging efficiency (typically 90–95%), $P_{charge}(t)$ = Charging power at time t (W), $P_{discharge}(t)$ = Discharging power at time t (W), C_{batt} = Total battery capacity (Wh), Δt = Time step (1 hour in this case).

Battery Operational Constraints [71]

$$SOC_{min} \leq SOC(t) \leq SOC_{max} \quad (6)$$

Charging occurs only when PV generation exceeds the load demand, while discharging occurs only when PV power is insufficient to meet the demand.

Battery Charging Power Equation [71]

$$P_{charge}(t) = \min \left(\max(0, P_{PV}(t) - P_{load}(t)), \frac{(SOC_{max} - SOC(t-1)) C_{batt}}{\eta_c \Delta t} \right) \quad (7)$$

Battery Discharging Power Equation [71]

$$P_{discharge}(t) = \min \left(\max(0, P_{load}(t) - P_{PV}(t)), \frac{(SOC(t-1) - SOC_{min}) C_{batt}}{\eta_d \Delta t} \right) \quad (8)$$

Battery Energy Balance Equation [72] Over the operational horizon, the battery must satisfy the energy balance condition-

Table 6

Cost information of hybrid renewable energy system components [13,66,74–76].

Characteristics	Solar Panel	Biogas	Converter	Battery
Model	EVO 6N SE6-66HBD	Generic Genset	CSI-1KTL1P-GI-FL	1 kWh Li-Ion
Power (W)	700	Auto-size	1000	1 kWh
Capital Cost (\$)	400	500 \$/kW	700	550
Replacement Cost (\$)	350	400 \$/kW	400	550
O&M Cost (\$/yr)	5	0.03 (\$/op. hour)	0	10

$$\sum_t (P_{\text{charge}}(t) \times \eta_c) = \sum_t \left(\frac{P_{\text{discharge}}(t)}{\eta_d} \right) \quad (9)$$

These equations ensure accurate modeling of the battery's charging/discharging dynamics, capacity limitations, and energy balance, thus allowing a realistic assessment of system reliability and performance.

2.4.4. Biogas generator

An auto-sized biogas generator is employed in this study as part of the hybrid renewable energy system. The auto-sizing feature allows HOMER software to dynamically determine the most appropriate generator capacity based on system load requirements, resource availability, and economic constraints. This flexibility enables HOMER to optimize system sizing more accurately, leading to improved cost-effectiveness and operational efficiency. The power output of the biogas generator, P_{BG} , is calculated using Equation (10) [73]:

$$P_{BG} = \frac{B_A \times CV_b \times \eta_g \times \eta_{gas}}{3600} \quad (10)$$

where B_A represents the amount of biomass available (kg/hr), CV_b refers to the calorific value of biogas (kJ/kg), η_g denotes the efficiency of the biogas generator (%), and η_{gas} represents the gasification efficiency of the gasifier (%). This formulation allows for accurate estimation of the generator's contribution to the hybrid system, ensuring that both cost and resource utilization are optimized.

Table 6 summarizes the economic parameters for each major system component. These cost assumptions, including capital, replacement, and O&M costs, serve as key inputs for evaluating the economic feasibility and guiding the optimization of the hybrid PV-biogas system.

2.5. Economic modeling

The economic analysis for this study employs HOMER software to evaluate the financial viability of the proposed hybrid renewable energy system. HOMER models and optimizes the system by considering capital investment, O&M costs, energy production, and revenue streams over the project's lifespan. This analysis evaluates several key financial indicators, NPC, LCOE, BEP, IRR.

Net present cost (NPC): The NPC represents the present value of all costs incurred over the lifetime of the system, discounted to present terms. HOMER Pro calculates NPC using Equation (11) [77]:

$$NPC = \sum_i^N \frac{NC_i}{(1+R)^i} - TInvC \quad (11)$$

where NC_i is the net cash flow in year i , R is the discount rate, and $TInvC$ is the total investment cost. A positive NPC indicates the project is financially attractive.

Internal rate of return (IRR): The IRR is a critical indicator of financial viability, representing the discount rate at which the NPC of all cash flows becomes zero. HOMER calculates IRR using Equation (12) [78]:

$$IRR = \sum_{i=0}^N \frac{NC_i}{(1+IRR)^i} \quad (12)$$

Table 7

Financial parameters associated with the project [21,56].

Parameter	Value
Discount rate (%)	8
Inflation rate (%)	7
Project lifetime (years)	25
Funding type	Self-funding
System type	On-grid and Off-grid
Grid power price (\$/kWh)	0.17
Grid net excess price (\$/kWh)	0.10

A higher IRR reflects greater profitability and a stronger return on investment.

Break-even point (BEP): The BEP determines how long it takes for the project to recover its initial investment. It is calculated using Equation (13) [78]:

$$BEP = \frac{TInvC}{CF} \quad (13)$$

where CF denotes the annual cash flow.

Table 7 summarizes the financial parameters used for the economic modeling of the hybrid system. These values form the basis for assessing the economic viability of the PV-biogas system and projecting the long-term financial performance of the investment.

2.6. Environmental impact modeling

In addition to financial considerations, this study evaluates the environmental benefits of the hybrid PV-biogas system. The focus is on quantifying greenhouse gas (GHG) emission reductions, a critical metric for assessing environmental sustainability. The GHG emission reduction rate is calculated using Equation (14) [79]:

$$\text{GHG Emission Rate} = \frac{TGHG(gCO_2)}{APG \times LT} \quad (14)$$

where $TGHG$ represents the total GHG emissions over the system's life-cycle (gCO_2), APG denotes the annual power generation (kWh), and LT is the system's lifetime (years). This modeling framework provides a clear understanding of the hybrid system's contribution to reducing carbon emissions and supports its alignment with global climate goals.

3. Result analysis

The Result Analysis section provides a comprehensive evaluation of the findings from both the HOMER simulations and the optimization algorithms. It emphasizes the performance, cost-efficiency, and environmental advantages of the proposed on-grid and off-grid PV-biogas hybrid systems tailored for Rosso, Mauritania.

3.1. HOMER optimization result

This section delivers an in-depth analysis of the simulation outcomes generated by the HOMER software. The focus is on evaluating different configurations of the PV-biogas hybrid system for both on-grid and off-grid applications in Mauritania. Key performance indicators such as energy production, system costs, and operational efficiency are assessed.

Architecture								Cost				System
				PV (KW)	Biogas Generator (kW)	Battery	Converter (kW)	NPC (\$)	COE (\$)	Operating cost (\$/yr.)	Initial Capital (\$)	Ren. Frac. (%)
				177	100	111	75.8	\$830,896	\$0.128	\$25,478	\$265,219	100
					100			\$1.13M	\$0.174	\$48,504	\$50,000	100
				4.17	100		0.708	\$1.13M	\$0.174	\$48,528	\$52,880	100
					100	5	0.708	\$1.13M	\$0.175	\$48,637	\$53,246	100

Fig. 11. HOMER Pro optimization result for off-grid system.

Table 8

Production summary of the off-grid optimal system.

Component	PV	Biogas	Total
Production (kWh/year)	329,964	156,689	486,653
AC Primary Load (kWh/year)	292,354		
Excess Electricity (kWh/year)	172,051		
Unmet Electric Load (kWh/year)	0		
Capacity Shortage (kWh/year)	0		

Table 9

PV system-specific information for the off-grid system.

Entry	Value
Rated Capacity	177 kW
Maximum Output	182 kW
Capital Cost	\$101,126
Replacement Cost	\$0
O&M Cost	\$28,065
Total Production	329,964 kWh/year
PV Penetration	113%
Hours of Operation	4,384 hours/year
COE	\$0.016/kWh

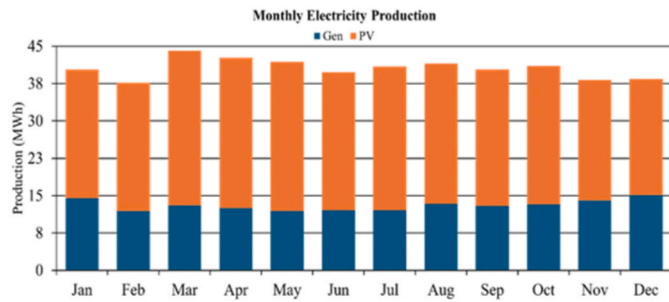


Fig. 12. Monthly electricity production for the off-grid system.

3.1.1. Off-grid system result analysis

Fig. 11 presents the optimization results from HOMER Pro for the most optimal off-grid system configuration, which includes 177 kW of PV capacity, a 100-kW biogas generator, 111 batteries, and a 75.8 kW converter. This configuration achieves a NPC of \$830,896, an LCOE of \$0.128, an annual operating cost of \$25,478, and an initial capital investment of \$265,219. It also boasts a 100% renewable fraction, indicating complete reliance on renewable energy sources. This optimal setup provides a cost-effective and sustainable solution for electricity generation, highlighting the feasibility of integrating biogas and PV systems for off-grid applications in Rosso, Mauritania.

Table 8 provides a production summary for the most optimal off-grid system configuration. The system produces a total of 486,653 kWh per year, with 329,964 kWh generated by the PV component and 156,689 kWh by the biogas generator. The system efficiently meets the AC primary load of 292,354 kWh per year without any unmet electric load or capacity shortage. Additionally, it generates 172,051 kWh of excess electricity annually, indicating a robust system capable of fully satisfying the energy needs and providing surplus power. Fig. 12 illustrates the Monthly Electricity Production for the off-grid system.

Table 9 provides specific information about the PV system within the optimal off-grid configuration. The PV system has a rated capacity of 177 kW and a maximum output of 182 kW. The capital cost is \$101,126, with no replacement cost. The O&M cost amounts to \$28,065. The total annual production is 329,964 kWh, achieving a PV penetration of 113%. The system operates for 4,384 hours per year, with an LCOE of \$0.016

Table 10

Biogas generator-specific information for Off-grid simulation results.

Quantity	Value	Units
Average Production	156,689	kWh/yr
Minimum Output	45.0	kW
Fuel Consumption	65.3	tons/yr
Specific Fuel Consumption	0.292	kg/kWh
Fuel Energy Input	69,840	kWh/yr
Average Electrical Efficiency	32.2	%
Number of Operation Hours	2,373	hrs/yr
Operational Lifetime	6.32	yr
Capacity Factor	17.9	%
Fixed Generation Cost	5.68	\$/hr
Marginal Generation Cost	0.000674	\$/kWh

per kWh. This data underscores the efficiency and cost-effectiveness of the PV component in the overall off-grid energy solution. Fig. 13 demonstrates the PV system output for the Off-grid system throughout the year.

Table 10 offers detailed information about the biogas generator within the Off-grid optimization result. The generator produces an average of 156,689 kWh per year, with a minimum output of 45.0 kW. It consumes 65.3 tons of fuel annually, with a specific fuel consumption of 0.292 kg/kWh and a fuel energy input of 69,840 kWh per year. The generator operates for 2,373 hours per year, with an average electrical efficiency of 32.2%. Its operational lifetime is estimated at 6.32 years, with a capacity factor of 17.9%. The fixed generation cost is \$5.68 per hour, while the marginal generation cost amounts to \$0.000674 per kWh. These parameters provide valuable insights into the performance and cost-effectiveness of the biogas generator component within the off-grid system configuration. Fig. 14 illustrates the Biogas generator output for the off-grid system.

Table 11 provides specific information about the battery system in the off-grid simulation result. The battery system has a rated capacity of 111 kWh and an annual throughput of 71,986 kWh per year. It incurs a maintenance cost of \$1,110 per year and provides an autonomy of

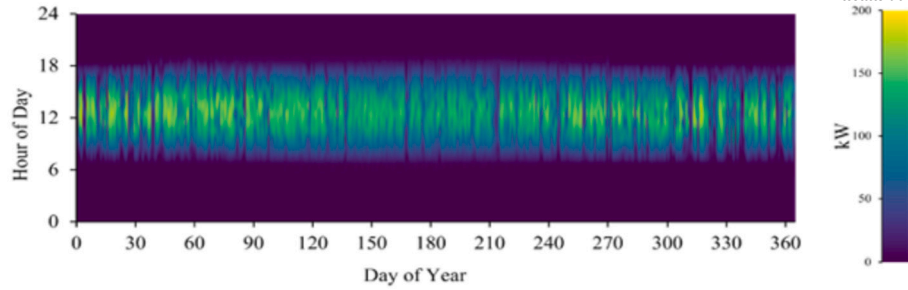


Fig. 13. PV system output for the off-grid system.

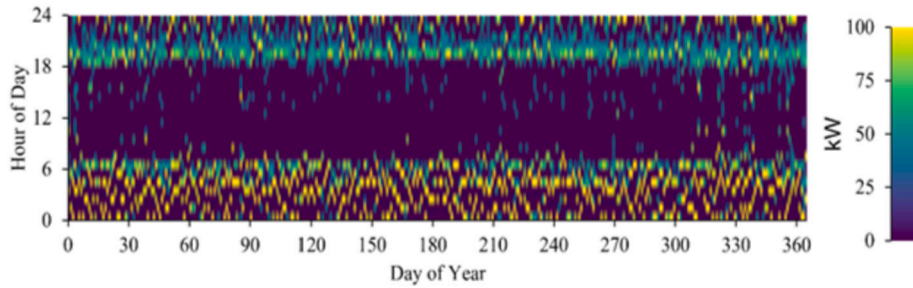


Fig. 14. Biogas generator output for the off-grid system.

Table 11

Battery system-specific information for Off-grid simulation results.

Entry	Value
Rated Capacity	111 kWh
Annual Throughput	71,986 kWh/yr
Maintenance Cost	\$1,110 /yr
Autonomy	2.66 hr
Expected Life	4.63 yr
Capital Costs	\$61,050
Losses	7,587 kWh/yr

Table 12

Production summary of the on-grid optimal system.

Production summary				
Component	PV	Biogas	Grid Purchase	Total
Production (kWh/year)	1,493,469	876,000	2,369,469	4,738,938
AC Primary Load (kWh/year)	292,354			
Grid Sales (kWh/year)	1,774,963			
Excess Electricity (kWh/year)	239,452			
Unmet Electric Load (kWh/year)	0			
Capacity Shortage (kWh/year)	0			

Table 13

PV system-specific information for the on-grid system.

Entry	Value
Rated Capacity	801 kW
Maximum Output	822 kW
Capital Cost	\$457,714.29
Replacement Cost	\$0
O&M Cost	\$127,029
Total Production	1,493,469 kWh/yr
Hours of Operation	4,384 hours/yr
LCOE	\$0.0160/kWh

2.66 hours. The expected lifespan of the battery system is 4.63 years. The capital cost for the battery system is \$61,050, with annual energy losses amounting to 7,587 kWh. These details highlight the role of the battery system in ensuring energy storage and reliability within the off-grid configuration. Fig. 15 depicts the battery systems output for the Off-grid system throughout the year.

3.2. On-grid optimized system result analysis

Fig. 16 presents the optimal on-grid system configuration, which features 801 kW of PV capacity, a 100 kW biogas generator, and a 408 kW converter, without a battery storage system. This configuration achieves a NPC of \$1.89 million, a LCOE of \$0.041, and an annual O&M cost of \$120,886. The initial capital investment amounts to \$793,314, with a renewable fraction of 100%. This setup demonstrates the feasibility of integrating biogas and PV systems into the on-grid network in Rosso, Mauritania, providing a sustainable and cost-effective solution for electricity generation.

Table 12 provides a production summary for the optimal on-grid system. The system generates 4,738,938 kWh annually, with 1,493,469 kWh from PV, 876,000 kWh from the biogas generator, and 2,369,469 kWh purchased from the grid. It meets the AC primary load of 292,354 kWh and sells 1,774,963 kWh of surplus electricity back to the grid each year. An additional 239,452 kWh of excess electricity is produced, with no unmet load or capacity shortage, indicating efficient energy utilization and surplus generation potential. Fig. 17 illustrates the monthly electricity production for the on-grid system.

Table 13 summarizes the PV system within the on-grid configuration. It has a rated capacity of 801 kW, peaking at 822 kW, with a capital cost of \$457,714.29 and no replacement cost. O&M expenses total \$127,029 annually. The system produces 1,493,469 kWh per year, operating for 4,384 hours, with an LCOE of \$0.0160/kWh. These figures highlight the PV system's efficiency and cost-effectiveness, underscoring its major contribution to electricity generation in Rosso. Fig. 18 shows the annual PV system output.

Table 14 details the biogas generator for the on-grid system. The generator is rated at 100 kW with an operational life of 1.71 years and a capital cost of \$50,000. It consumes 341 tons of fuel annually, operating 8,760 hours per year at a fixed generation cost of \$5.68/hr. Biogas is used as the sole fuel at \$1.00/kg, with annual maintenance costs of \$26,280. The generator produces 876,000 kWh per year, achieving a marginal generation cost of just \$0.000674/kWh.

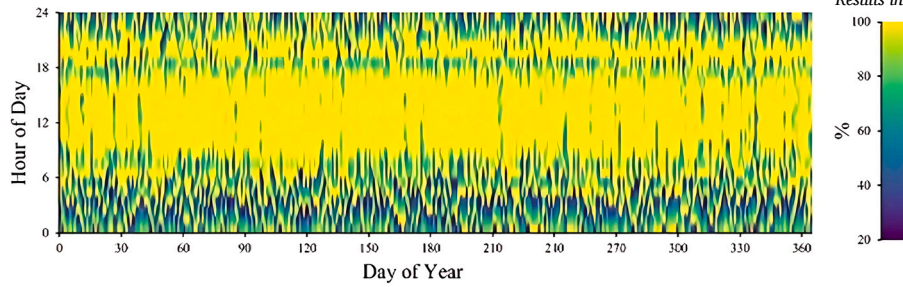


Fig. 15. Battery system output for the off-grid system.

Architecture										Cost		System	
					PV (KW)	Biogas Generator (kW)	Battery	Converter (kW)	NPC (\$)	COE (\$)	Operating cost (\$/yr.)	Initial Capital (\$)	Ren. Frac. (%)
					801	100		408	\$1.89M	\$0.041	\$120,886	\$793,314	100
					801	100	2	408	\$1.89M	\$0.041	\$120,777	\$793,152	100
						100			\$161,905	\$0.008	\$9,544	\$50,000	100
						100	1		\$160,716	\$0.008	\$9,518	\$50,612	100

Fig. 16. HOMER Pro optimization results for the on-grid system.

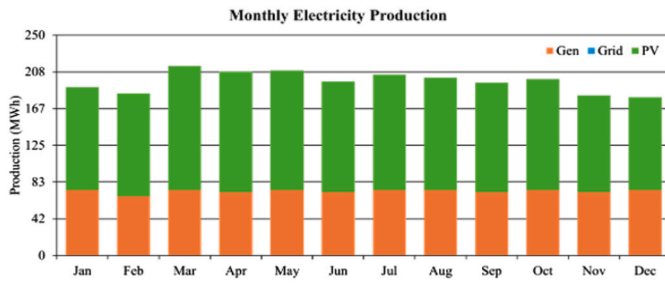


Fig. 17. Monthly electricity production for the on-grid system.

Table 14

Biogas generator-specific information for the on-grid simulation results.

Entry	Value
Capacity	100 kW
Operational Life	1.71 yr
Capital Cost	\$50,000
Fuel Consumption	341 tons/yr
Hours of Operation	8,760 hrs/yr
Fixed Generation Cost	\$5.68/hr
Generator Fuel	Biogas
Fuel Price	\$1.00/kg
Maintenance Cost	\$26,280/yr
Electrical Production	876,000 kWh/yr
Marginal Generation Cost	\$0.000674/kWh

3.3. Economic analysis

The Economic Analysis section explores the financial aspects of the proposed PV-biogas hybrid systems in Rosso, Mauritania using the HOMER platform. This assessment evaluates the economic feasibility, cost-effectiveness, and overall financial viability of both on-grid and off-grid configurations.

Table 15 summarizes the costs associated with both the proposed on-grid and off-grid systems. The table categorizes costs into capital, replacement, O&M, and salvage costs for the various components of each configuration. For the off-grid system, the biogas generator incurs a capital cost of \$50,000, a replacement cost of \$106,809, and an O&M cost

of \$158,058, resulting in a total of \$316,340. Likewise, the PV system, battery, and converter each contribute their respective costs, culminating in a total system expenditure of \$1,890,619. In contrast, the on-grid system demonstrates different cost dynamics, particularly with the grid component, which significantly influences the total cost structure due to its reliance on external infrastructure.

Table 16 highlights the key financial metrics calculated by HOMER for both the on-grid and off-grid PV-biogas systems, using equations (6), (7), and (8). The analysis reveals that the on-grid system offers a significantly lower LCOE at \$0.041 per kWh, compared to \$0.128 per kWh for the off-grid system. Additionally, the on-grid system achieves a faster break-even point at 6.6 years, whereas the off-grid system takes 8.54 years. The IRR for the on-grid system stands at 14%, outperforming the 9.9% IRR of the off-grid configuration. Similarly, the ROI is higher for the on-grid system at 11%, in contrast to 6.7% for the off-grid option. These results indicate that the on-grid system is financially more viable and cost-effective for implementation in Rosso, Mauritania.

3.4. Environmental impact assessment

The Environmental Impact Assessment section aims to evaluate the ecological implications of implementing on-grid and off-grid systems in Rosso, Mauritania. Table 18 provides a comparative analysis of GHG emissions between the two system configurations. In the off-grid system, carbon dioxide emissions are recorded at 10.3 kg/year, while the on-grid system exhibits higher emissions at 53.7 kg/year.

Using the data provided in Table 17 and applying Equation (4), this study calculated the total emissions of CO_2 for both the on-grid and off-grid systems. For the on-grid system, with an emissions value of 53.7 kg/year of CO_2 , and assuming that all electricity consumed from the grid in Mauritania is produced by this system, the total emissions would be approximately 1,113 tons of CO_2 per year. Similarly, for the off-grid system, with an emissions value of 10.3 kg/year of CO_2 , the total emissions would be around 229 tons of CO_2 per year under the same assumption. This analysis underscores the significant difference in carbon emissions between the two system configurations and highlights the potential environmental benefits of implementing off-grid renewable energy solutions.

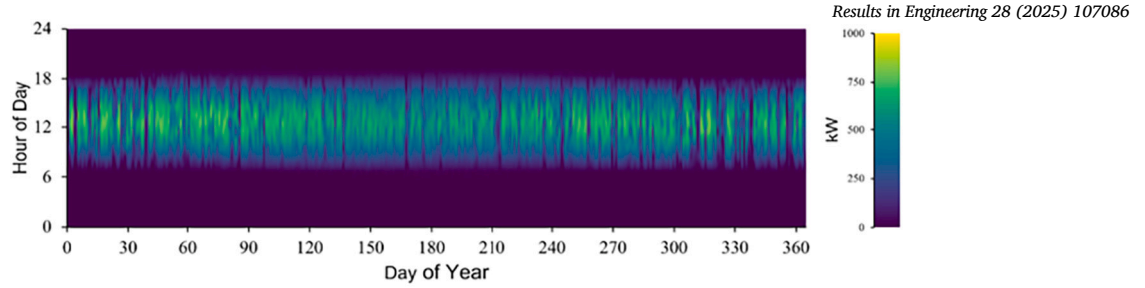


Fig. 18. PV system output for the on-grid system.

Table 15

Cost summary of the proposed on-grid and off-grid systems.

Cost Type	Capital (\$)		Replacement (\$)		O&M (\$)		Salvage (\$)		Total (\$)	
Components	Off-grid	On-grid	Off-grid	On-grid	Off-grid	On-grid	Off-grid	On-grid	Off-grid	On-grid
Biogas Generator	50,000	50,000	106,809	497,966	158,058	583,475	-1,427	-12,680	316,340	1,133,920
PV System	101,126	457,714	0.00	0.00	28,065	127,029	-11,688	-52,899	117,504	531,843
Battery	61,050	–	268,778	–	24,645	–	-28,819	–	325,653	–
Converter	53,042	285,600	26,363	141,945	0.00	0.00	-8,007	-43,112	71,398	384,433
Grid	–	0.00	–	0.00	–	-3,940,816	–	0.00	–	-3,940,816
System		793,314		639,911		-3,230,312		-108,692		-1,890,619

Table 16

Financial metrics for the on-grid and off-grid systems calculated by HOMER.

Entry	Economic Metrics	
	On-grid	Off-grid
Levelized Cost of Energy (LCOE) (\$/kWh)	0.041	0.128
Break-even point (years)	6.6	8.54
Internal Rate of Return (IRR) (%)	14	9.9
Return on Investment (ROI) (%)	11	6.7

Table 17

GHG emission data for the on-grid and off-grid systems.

Pollutants Name	Emissions (kg/year)	
	Off-grid System	On-grid system
Carbon Dioxide	10.3	53.7
Carbon Monoxide	1.08	5.63
Unburned Hydrocarbons	0.0470	0.246
Particulate Matter	0.00653	0.0341
Sulfur Dioxide	0	0
Nitrogen Oxides	1.01	5.29

3.5. Optimization algorithm result

Table 18 presents the optimized results obtained from the GWO, WOA, and HOMER software for the PV-Biogas hybrid system. Each algorithm provides a configuration for system components, along with key financial and performance metrics, including the LCOE, NPC, and PP.

HOMER identifies an optimal configuration of 801 kW PV, 100 kW BG, and a 408 kW converter, achieving an LCOE of \$0.041/kWh, NPC of \$1.89M, and a PP of 6.6 years. GWO slightly refines these metrics, lowering LCOE to \$0.038/kWh and NPC to \$1.81M, with a PP of 6.4 years, using 793 kW PV and 98 kW BG. WOA yields the best result, further reducing LCOE to \$0.036/kWh, NPC to \$1.77M, and PP to 6.1 years, with 781 kW PV and 98 kW BG. Although GWO and WOA deliver marginal gains in system sizing and economics, the improvements are minor and do not substantially diverge from HOMER's outcome. Instead, they fine-tune the configuration, confirming that HOMER's solution remains strong while benefiting from slight optimizations from these algorithms.

4. Comparison with other studies

This section critically compares the performance and economic outcomes of the proposed PV–BG hybrid system for Mauritania with selected HRES implementations reported in recent literature. The comparison focuses on key indicators, LCOE, NPC, and RF, to contextualize the findings of this study within the broader landscape of hybrid system research.

Table 19 and Table 20 present a comparative analysis of HRES implemented across different regions, with a clear separation between on-grid and off-grid systems. Both tables highlight critical techno-economic indicators, namely COE, NPC, and RF, illustrating the diversity in system configurations and performance outcomes. For on-grid systems, COE values range from 0.041\$/kWh (Brazil and Mauritania) to 0.240\$/kWh (Egypt), with variations primarily driven by hybrid mix, local resource availability, and reliance on grid support. The proposed on-grid PV–Biogas system in Mauritania achieves one of the lowest COE values at 0.041\$/kWh, while maintaining a 100% renewable fraction and an NPC of 1.89MUSD, underscoring its strong techno-economic feasibility compared to global benchmarks.

In contrast, off-grid systems exhibit wider variability, with COE ranging from 0.0712\$/kWh (India) and 0.0728\$/kWh (Pakistan) to as high as 0.478\$/kWh (Peru). The NPC values also span from small-scale community systems to multi-million-dollar projects. Importantly, the off-grid PV–Biogas configuration developed in this study for Mauritania achieves a COE of 0.128\$/kWh with a 100% renewable fraction, outperforming several comparable systems. Furthermore, when compared to advanced hybrid configurations such as the PV–WT–DG–BESS–FC–ELZ–HT system reported for Egypt [71], our proposed solution demonstrates competitive economic performance while employing a simpler and more locally adaptable design. Overall, the results confirm that the dual on-grid and off-grid configurations proposed in this study provide cost-effective and fully renewable energy supply options. This dual perspective enhances the robustness of the analysis, offering practical pathways for sustainable electrification in developing regions with varying levels of grid access.

5. Discussion

The findings of this study underscore the considerable potential of HRES, particularly PV–biogas configurations, in addressing electricity generation challenges not only in Mauritania but also in other regions

Table 18
Optimized results by HOMER and metaheuristic algorithms.

Optimization Algorithm	PV (kW)	BG (kW)	Battery	Converter (kW)	LCOE (\$/kWh)	NPC (\$)	PP (years)
HOMER	801	100	-	408	0.041	1.89M	6.6
GWO	793	98	2	399	0.038	1.81M	6.4
WOA	781	98	-	388	0.036	1.77M	6.1

Table 19
Comparative analysis of On-grid system HRES in different regions.

Year	Country	Hybrid system	COE (\$/kWh)	NPC (\$)	RF (%)	Ref
2024	Egypt	PV/Wind/Battery/DG/Grid	0.240	533,654	78	[31]
2022	Jordan	PV/Biogas/Grid	0.23	\$54.5	–	[40]
2024	Brazil	PV/Biogas/Grid	0.0413	–	–	[50]
2020	Himalaya	PV/BM/Grid	0.102	42,081	83	[80]
2025	China	PV/BDG/Battery/Grid	0.082	1.04M	59.9	[81]
2025	Mauritania	PV/BG/Grid	0.041	\$1.89M	100	This Study

Table 20
Comparative analysis of Off-grid system HRES in different regions.

Year	Country	Hybrid system	COE (\$/kWh)	NPC (\$)	RF (%)	Ref.
2022	Ghana	PV/BG/Battery	0.256	506,629	100	[23]
2023	Ethiopia	PV/BG/Bio-diesel	0.175	273,887	64	[82]
2021	Peru	PV/Wind/Diesel	0.478	227,335	94	[83]
2020	Nigeria	PV/Diesel/Battery	0.106	1M	77.4	[75]
2025	Kenya	PV/Biogas/Battery	0.171	183,558	–	[84]
2024	Pakistan	PV/BG/Hydro/Battery	0.0728	\$152,242	–	[36]
2025	Turkey	PV/Biogas/Hydrogen	0.2094	\$1.9M	–	[37]
2023	Egypt	PV/Wind/Biomass/Hydro	0.2750	\$8.05M	–	[41]
2022	Egypt	PV/BG/Battery	0.0801	\$3.3M	–	[42]
2022	China	PV/Biogas/Diesel/Battery	0.24	\$1.81M	–	[44]
2024	Morocco	PV/Biogas/Battery	0.280	–	–	[45]
2023	India	PV/Biomass/Battery	0.0712	\$159,648	–	[46]
2025	Cameroon	PV/Biogas/Battery/DG	0.07563	\$300,661	–	[47]
2024	Türkiye	Biogas/PV/Wind	0.0688	\$70,777	89.9	[49]
2025	Egypt	PV/WT/DG/BESS/FC/ELZ/HT	0.153	1.78M	–	[71]
2025	Mauritania	PV/BG	0.128	\$830,896	100	This Study

with similar climatic and socio-economic contexts. Although this study focuses on the specific location of Rosso, Mauritania, the broader applicability is justified by the commonality of key resource characteristics such as high solar irradiance, abundant biomass residues, and energy access challenges prevalent across many developing countries in West Africa and beyond. Therefore, the insights derived here can inform decentralized energy planning and hybrid system deployment strategies in comparable rural and peri-urban settings worldwide, contributing to global efforts in sustainable development and energy poverty alleviation. The integration of both on-grid and off-grid systems in this study highlights versatile pathways for improving energy access. The optimized off-grid system, identified through HOMER Pro, includes a combination of 177 kW PV, 100 kW biogas generator, 111 batteries, and a 75.8 kW converter. This configuration achieves a NPC of \$830,896, a CoE of \$0.128/kWh, and demonstrates operational feasibility with an annual PV generation of over 1.4 million kWh. Conversely, the on-grid system shows a lower capital cost and a remarkably low LCOE of \$0.041/kWh, indicating high cost-effectiveness in grid-connected contexts. Collectively, these results suggest that hybrid PV–BG systems can offer flexible solutions adaptable to different infrastructural conditions.

Regarding concerns about the reliability and operational challenges associated with biogas generators, it is recognized that such equipment is relatively large and may introduce maintenance complexities. To mitigate these risks, this study incorporates component replacement costs and operational lifetimes into the techno-economic analysis, ensuring realistic lifecycle considerations. Moreover, the use of batteries and converters as part of the hybrid system provides operational buffering and load management, enhancing system resilience. Future work could focus on detailed reliability modeling and maintenance schedul-

ing to further improve system robustness. The methodological approach in this study combines HOMER Pro for baseline system modeling with advanced metaheuristic algorithms—Grey Wolf Optimizer (GWO) and Whale Optimization Algorithm (WOA)—for solution refinement. This dual-step process leverages the proven reliability of HOMER Pro's simulation framework with the global search and exploitation capabilities of GWO and WOA, ensuring both computational efficiency and high-quality convergence. While numerous other optimization algorithms, such as Mayfly, Constrained Particle Swarm Optimization (CPSO), Harmony Search (HS), and Flower Pollination Algorithm (FPA), have been successfully applied in similar research domains, it is neither computationally practical nor methodologically necessary to implement an exhaustive list of all available algorithms within a single study. Our selection of GWO and WOA was based on their complementary search behaviors, track record in renewable energy system optimization, and suitability for the problem scale. The focus was on achieving a robust and well-validated solution within the study scope, rather than conducting a comparative algorithmic survey. Nevertheless, future work could explore additional algorithms to further test the generalizability and sensitivity of the obtained results. The economic and environmental metrics presented here align well with prior studies in sub-Saharan Africa and South Asia, emphasizing the potential of hybrid renewable energy systems to reduce greenhouse gas emissions and reliance on imported fossil fuels. Importantly, this study extends previous research by concurrently evaluating both on-grid and off-grid configurations, providing a holistic perspective on feasible deployment pathways under varying infrastructural conditions. Such comprehensive analysis supports informed policymaking and investment decisions.

5.1. Policy recommendations

In light of the encouraging results, several policy measures are recommended to accelerate the adoption of HRES in Mauritania and similar regions. Governments should consider fiscal incentives such as tax breaks and subsidies tailored to renewable energy infrastructure. Regulatory frameworks that streamline project approval and incentivize private sector participation will be spatial to scaling deployment. Supporting research and development, particularly in improving PV integration and biogas production technologies, can enhance system performance and cost-effectiveness. Partnerships with international organizations and technology providers can facilitate capacity building and technology transfer. Furthermore, formulating a national renewable energy strategy with clear targets will provide direction and foster investor confidence. Together, these measures can catalyze a sustainable energy transition that balances economic growth with environmental stewardship.

5.2. Limitations and future scope

This study acknowledges several limitations. The analysis relies on simulation models that, while comprehensive, may not capture all real-world uncertainties such as equipment degradation, fluctuating market prices, and communication system failures. Notably, degradation effects for PV panels, battery state-of-charge dynamics, and component replacement schedules were incorporated to improve realism; however, further work is needed to model operational contingencies and maintenance impacts more explicitly. Additionally, the study does not explicitly address grid stability challenges arising from high renewable penetration in the on-grid scenario, which should be a focus of future research. Social, cultural, and behavioral factors influencing technology adoption were beyond the current scope but remain significant to successful implementation and warrant interdisciplinary investigation. Lastly, the environmental analysis was limited to greenhouse gas emissions; a more comprehensive life cycle assessment including land use, water consumption, and biodiversity impacts would provide a fuller sustainability evaluation. In short, this work offers a robust assessment of hybrid PV–biogas systems tailored to Mauritania's context, with broader implications for similar settings. By leveraging advanced optimization techniques and considering both grid-connected and standalone configurations, it lays a foundation for informed decision-making towards sustainable and resilient energy systems.

6. Conclusion

This study delivers a detailed assessment of hybrid PV–BG systems in Mauritania, affirming its economic and environmental viability. The findings reveal that the on-grid configuration is the most cost-effective, achieving an LCOE of \$0.041/kWh, a shorter payback period of 6.6 years, and superior financial returns (IRR 14% and ROI 11%). In comparison, the off-grid system, though having a higher LCOE of \$0.128/kWh, provides complete energy independence with a 100% renewable fraction and reaches an NPC of \$830,896, making it particularly suitable for remote areas with limited grid access. Both systems significantly curb GHG emissions and reliance on fossil fuels, advancing the vision of a more sustainable energy landscape. The analysis demonstrates that PV–BG hybrids can serve as a viable solution for energy-deficient regions, enhancing energy security while offering long-term economic benefits. Nonetheless, several limitations must be acknowledged. The study relies on simulation-based modeling, which may not fully capture real-world operational complexities. It does not account for broader ecological or social factors, nor does it assess potential grid stability issues for the on-grid scenario under high RES penetration. Additionally, the scope is geographically narrow, focusing exclusively on Mauritania. Future work should explore alternative hybrid setups such as wind–BG or hydro–BG to diversify insights and evaluate a wider range

of environmental parameters, including land use, water demand, and ecosystem impacts. Moreover, incorporating economic considerations like policy incentives, carbon credits, and subsidy schemes would offer a deeper understanding of the financial attractiveness of these systems. Such efforts will refine hybrid renewable energy models and facilitate their scalable deployment across similar regions worldwide.

CRedit authorship contribution statement

Sid Ahmed Sidi Habib: Writing – original draft, Validation, Resources, Conceptualization. **Md. Ashraful Islam:** Writing – review & editing, Writing – original draft, Visualization, Software, Methodology. **M.M. Naushad Ali:** Writing – review & editing, Supervision, Investigation, Formal analysis, Conceptualization. **Ian B. Benitez:** Visualization, Methodology, Data curation. **Zokir Mamadiyarov:** Resources, Methodology. **Bobur Mirzayev:** Visualization, Data curation. **Hayitov Abdulla Nurmatovich:** Methodology. **Aymen Flah:** Resources, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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